

Navigators Technical Approach

Modular Architectural Specifications, State Machine & Offline Vector Tile Engine

Navigators Technical Architecture

Technical Stack: Python 3.14, PyTorch, FastAPI, ONNX Runtime, JavaScript (ES2022)
Database: SQLite (WAL mode) / PostgreSQL

Technology Stack and Component Blueprint

Navigators utilizes a decoupled technical architecture optimized for high performance, strict typing, and cross-platform deployment.

1. Machine Learning and Inference

- **PyTorch:** Deep learning framework used for training specialized Temporal Convolutional Networks (TCN) and ResNet1D velocity estimation models.
- **ONNX Runtime (Python & Web):** Standardized cross-platform inference framework enabling seamless execution of trained models across Python services and client-side browser runtimes (*onnxruntime-web*).

2. Core Navigation and Mathematics

- **NumPy & SciPy:** C-accelerated array processing, linear algebra, vector projections, and matrix operations for the 15-state Extended Kalman Filter (EKF).
- **Client-Side Engine:** Pure JavaScript ES2022 implementation (*simulator/offline_engine.js*) providing lightweight, zero-dependency browser navigation.

3. Backend Services and API

- **FastAPI:** Asynchronous Python web framework delivering high-throughput REST endpoints, OpenAPI schemas, and strict request/response data validation.
- **Uvicorn:** ASGI web server running FastAPI backend processes.
- **Pydantic v2:** Data validation and serialization layer defining immutable API contracts.

4. Database and Storage Layer

- **SQLite / PostgreSQL:** Relational storage tier managing 19 database tables, foreign key constraints, session tokens, audit trails, dataset registries, and model candidate metadata.

Coordinate Transformations and Pipeline Overview

Navigators uses four primary coordinate reference frames:

- **Sensor Body Frame (B):** Smartphone physical accelerometer and gyroscope axes.
- **Vehicle Kinematic Frame (V):** Forward (longitudinal), Right (lateral), and Down/Up (vertical) vehicle body directions.
- **Local East-North-Up Frame (ENU):** Tangent-plane Cartesian coordinate system centered at local navigation origin.
- **Geodetic Frame (WGS84):** Global Latitude, Longitude, and Ellipsoidal Altitude coordinates.

```
flowchart LR
    IMU["Sensor IMU (Frame B)"] --> Alignment["Triad Frame Alignment"]
    Alignment --> Kinematic["Vehicle/Pedestrian (Frame V)"]
    Kinematic --> TCN["Deep TCN Velocity Regressor"]
    TCN --> EKF["15-State Extended Kalman Filter"]
    EKF --> ENU["Local Coordinates (Frame ENU)"]
    ENU --> WGS84["Geodetic Conversion (WGS84)"]
```